

A Free Lunch? Adapting PP-OCRv6 for Historical Text Recognition

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Abstract

Despite impressive reported scores, large vision-language models have seen limited practical uptake in historical automatic text recognition because of their computational cost, dependence on large-scale pretraining, and hallucination. Historical ATR therefore continues to rely largely on compact CRNN line recognizers, which are visually grounded and trainable on modest data. Lightweight recurrence-free recognizers promise the accuracy of larger models with the practical advantages of CRNNs, yet have not been comprehensively evaluated on historical writing. We adapt PP-OCRv6, a recent compact text recognizer without strong language modeling, for historical line recognition and compare it with a conventional CRNN across generalized pretraining, domain-specific training, corpus-level fine-tuning, and manuscript-specific few-shot adaptation on multilingual Latin- and Arabic-script material. While PP-OCRv6 does not consistently outperform the baseline when trained from scratch, heterogeneous pretraining produces markedly better generalization. Comparisons with the Qwen3.5-based Medusa recognizer further show that fine-tuned PP-OCRv6 can outperform a large VLM tailored towards historical Latin-script HTR.

Keywords: automatic text recognition, historical documents, handwritten text recognition, optical character recognition, PP-OCRv6

1 Introduction

For the better part of a decade, line-wise recognition with convolutional recurrent neural networks (CRNNs) trained using connectionist temporal classification (CTC) [15] has been the dominant paradigm in automatic text recognition (ATR), including for historical material. Typically combining a convolutional feature extractor, bidirectional long short-term memory layers, and a linear projection [23; 27], these systems are accurate, robust, trainable on the modest amounts of annotated data available for individual historical corpora, and require no character-level segmentation. For these reasons, they underlie widely used recognition engines such as Calamari [33], Tesseract 4 [31], and kraken [18].

Text recognition has recently shifted towards large vision-language models. Besides achieving excellent error rates in conventional line recognition, these models can recognize entire paragraphs or pages without explicit segmentation. Unlike earlier methods such as Start, Follow, Read [30], which learn line extraction and recognition jointly, Qwen-VL [2] and LightOnOCR [29] directly produce ordered text and document structure from complete page images. Dispensing with line segmentation is especially attractive for historical ATR, where segmentation errors are frequently responsible for a significant proportion of overall recognition errors.

Generative recognition, in particular in the full-page setting, has nevertheless seen limited adoption in practical historical ATR. Autoregressive systems with strong language

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models can produce insufficiently grounded transcriptions, deriving text from linguistic priors without corresponding visual evidence. This and similar phenomena, commonly called hallucination, are still poorly understood. Recent work on Ancient Greek editions has demonstrated extensive overcorrection even in the presence of extensive changes such as word shuffling [17], pointing towards a fundamental problem in the construction of these systems.

Large autoregressive models are also expensive to train and deploy, and acquire their capabilities from pretraining corpora that cannot reasonably be produced for most historical collections. The mismatch is especially acute for non-Western traditions, obsolete scripts, minority and low-resource languages, and projects using specialized transcription norms. Synthetic data can alleviate annotation scarcity, for example by rendering transcriptions in digital typefaces approximating the target hands [4]. This approach remains limited to writing with suitable typographic surrogates, which places substantial portions of the written record beyond the practical reach of models that rely on large-scale pre-training.

These drawbacks have renewed interest in compact recognizers that replace recurrent layers without introducing high-capacity generative language models. Such systems retain the modest data requirements and visual grounding of conventional line recognition while permitting parallel execution on modern hardware. PP-OCRv6 is a recent general-purpose example whose authors report strong results on printed and handwritten text in document scans and natural scenes, with models ranging from 1.1 to 19 million parameters [36].

This article asks whether PP-OCRv6 offers a “free lunch” for historical ATR as a replacement for recurrent line recognition. The method is adapted for historical line images and evaluated when trained from scratch on project-specific data, pretrained as a generalized model, and fine-tuned towards particular datasets and transcription norms. Comparisons with kraken and Medusa, a Qwen3.5-9B-based line recognizer [22], examine recognition accuracy, few-shot adaptation, and inference throughput.

2 Related Work

2.1 Recurrent line recognition with CTC

Connectionist temporal classification permits alignment-free sequence modelling and therefore permits training from line images and their transcriptions without character-level annotations [15]. Combined with a convolutional feature extractor and bidirectional LSTMs, it has defined historical machine-printed and handwritten text recognition for the last decade; representative examples include the systems described by [23] and [33]. Although highly accurate, their recurrent sequence processing cannot fully exploit the hardware and software optimizations available to convolutional and attentional models.

2.2 Language-generative recognition

The principal attraction of language-generative recognition is its promise of higher recognition accuracy through the combination of visual features and strong pretrained language models. Representative systems include TrOCR, which combines pretrained image and text Transformers [20], and DTrOCR, whose decoder is initialized from GPT-2 [12]; both operate on presegmented text images.

Generative models also offer greater flexibility with respect to segmentation. Removing line segmentation as a fragile component of the recognition pipeline has been a longstanding objective. Start, Follow, Read learns line localization and recognition jointly [30],

OrigamiNet unfolds the two-dimensional page into a sequence [34], and DAN uses attention to traverse the document [9]. These pre-VLM systems reconstruct an internal segmentation through task-specific architectures that are brittle and difficult to train. Large VLMs instead use a common autoregressive formulation across text lines, paragraphs, and complete pages. Qwen-VL provides this flexibility as part of a general-purpose prompted model [2]; LightOnOCR applies it specifically to complete-page recognition [29]; and Churro is designed specifically for full-page historical ATR [25].

2.3 Compact recurrence-free recognizers

Attempts to eliminate recurrence long predate the widespread adoption of large Transformer models. [28] explored fully convolutional sequence decoding, while [13] and [35] retained CTC but replaced recurrent sequence modelling with convolutions. Despite strong in-domain results, these models often generalized less well than recurrent systems in practical transfer. This remained obscured by limited in-domain evaluation, which is a poor predictor of performance under domain shift [14].

More recent recurrence-free recognizers include the convolutional Gated Fully Convolutional Network [8] and Easter2.0 [5]. Attentional designs include the Modular Light Transformer [3], HTR-VT [21], and SVTRv2 [11]. Their scale remains far below that of language-generative recognizers: the Modular Light Transformer has 7.7 million parameters [3], HTR-VT 53.5 million [21], and PP-OCRv6 1.1-19 million, compared with one billion for LightOnOCR, itself considered a lightweight VLM.

3 Methodology

PP-OCRv6 is a two-stage OCR system for document scans and natural scene images. As only its potential for replacing a CRNN baseline is evaluated, only its recognition stage is considered.

3.1 PP-OCRv6 text recognition

The recognition component combines an LNetV4 convolutional backbone, an attentional LightSVTR neck, and a CTC head. The tiny variant omits LightSVTR, while the small and medium variants increase backbone capacity and neck width. The generalized recognizers reported by the authors support Chinese and contain 1.1, 5.2, and 19 million parameters. The smaller character inventories used in our experiments reduce the size of the final linear projections and the respective parameter counts to 0.68, 3.22, and 15.86 million.

Training combines equally weighted CTC and label-smoothed cross-entropy from an auxiliary autoregressive NRTR decoder [26]. The decoder acts as a regularizer and is discarded for inference, which remains CTC-based. The original method uses the Adam optimizer and a canonical input size of 48×320 pixels.

PP-OCRv6 inherits a data curation procedure from PP-OCRv5 [10] intended to filter noisy, incorrect, and trivially easy transcriptions. A referee recognizer assigns each line in the training dataset a mean character confidence. Samples below 0.80 are treated as difficult or noisy and those above 0.97 as trivial; around half of the original training set falls within the 0.95-0.97 confidence range the authors determine to be optimal.

3.2 Adaptation to historical text recognition

The nominal PP-OCRv6 input line height is doubled from 48 to 96 pixels to preserve fine strokes, diacritics, and abbreviation marks in degraded historical text. In preliminary experiments, further increases did not result in additional validation score gains. This intervention is consistent with other systems optimized for historical ATR: 128 pixels in PyLaia [24] and 120 in kraken [18], compared with 36 pixels in Tesseract 4 [32], which is optimized for modern machine-printed text.

The default PP-OCRv6 configuration imposes two restrictive limits that are manifestly unsuitable for historical ATR: NRTR targets are capped at 23 code points, and every line image is resized to 48×320 pixels. At the model’s horizontal stride of eight, the latter provides only 40 CTC steps, which is insufficient for historical lines of even moderate length. Lines are instead scaled to 96 pixels in height while preserving their aspect ratio, so that the available CTC sequence length grows with the input line. The NRTR target limit is raised accordingly.

The distillation loss using a medium teacher model for the original PP-OCRv6 tiny variant is omitted.

Confidence-window filtering is also discarded. Its lower boundary was tested on the French, Latin, and Spanish subset of the generalized-model training corpus, where it rejects 17% of lines, compared with approximately 3% of the original training data [10]. The medium-variant generalized model obtains a CER of at most 15% on 52% of the rejected material. The correctly annotated examples in Figure 1 receive filter confidences between 0.55 and 0.68 despite being reproduced exactly by the generalized model.

Confidence alone is therefore insufficient for detecting incorrect annotations because domain familiarity and annotation quality are intermingled in a single score. On heterogeneous historical data, confidence filtering removes difficult but correctly annotated lines and underrepresents rare scripts, hands, and typographies. It is therefore not used for pretraining.

The Adam optimizer was replaced with a hybrid optimizer, applying Muon [16] to hidden linear weights and AdamW to all remaining parameters, initially with the aim of accelerating convergence.

3.3 Experiments

The adapted tiny, small, and medium PP-OCRv6 variants are evaluated under four scenarios: **generalized-model training** for general-purpose, off-the-shelf recognition; **training from scratch** on a domain-specific corpus when no suitable base model is available; **fine-tuning** a generalized model for the best project-specific accuracy; and **few-shot adaptation** for the highest possible accuracy on a single manuscript.

They are compared with the default CRNN recognizer of the kraken ATR software [18], a four-layer convolutional network followed by three 200-unit bidirectional LSTM layers trained with CTC. Training was performed on NVIDIA A40 GPUs,

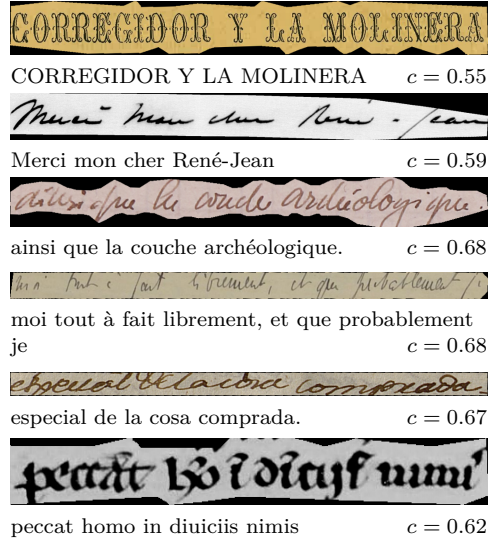


Figure 1: Correctly annotated lines rejected by confidence-window filtering.

using two devices for the medium from-scratch runs and one otherwise.

The generalized models were pretrained on a heterogeneous corpus combining hand-written, typed, and machine-printed in 37 different languages (ca. 6.2 million lines), and synthetic material (ca. one million lines).¹ Its held-out test split contains approximately 83000 lines in 33 languages. Almost all CATMuS and MAKHZAN training material is included in the pretraining corpus, while their evaluation sets remain held out.

The principal domain-specific evaluation uses CATMuS Medieval [7]. After removal of 7000 lines for validation, approximately 142000 lines from 300 manuscripts remain. CATMuS is relatively large for a historical ATR dataset and uniform domain-specific datasets are frequently substantially smaller. The training lines are therefore divided into nested 25, 50, 75, 90, and 100% tiers, each retaining all manuscripts, emulating these smaller training corpora.

For from-scratch training, each model is trained for a number of steps equivalent to 20 epochs on the complete corpus; the number of epochs is increased inversely with tier size so that all runs process the same number of line images. Fine-tuning is performed for eight epochs. Evaluation uses the ICDAR 2026 Competition on Multilingual Medieval Handwriting Recognition (CMMHWR) test set [19]: French, Latin, and Spanish test transfer to languages represented in CATMuS (task 1), Occitan tests transfer within the Romance family (task 2), and Czech tests cross-family transfer (task 3). A randomly sampled test set used for quality estimation during the creation of the CoMMA corpus [6]² provides an additional, more distant evaluation on medieval Latin and French material.

Accuracy is also compared with Medusa, a Qwen3.5-9B line recognizer submitted at CMMHWR [22], as an example of the recognition capabilities of current large VLMs. Medusa is evaluated through the authors’ DocWorkflow pipeline, including its CATMuS-specific post-processing. VLMs often struggle with historical hands and diplomatic transcription norms in zero-shot settings, therefore a system that has been tuned towards the evaluated corpora provides a fairer comparison.

To test whether the observed transfer behaviour is specific to Latin script, e.g., due to a larger and more diverse sample of Latin-script material in the pretraining dataset, zero-shot and fine-tuning evaluation is repeated on the Arabic, Persian, and Ottoman Turkish subsets of OpenITI MAKHZAN [1], a corpus of historical handwritten Arabic-script material. In contrast to the medieval Latin-script evaluation, the test set is not document-disjoint due to the construction of the pretraining corpus.

Few-shot manuscript-specific adaptation was evaluated separately on four CMMHWR test pages using between 20 and 240 training lines. Each page was divided into a training pool and a fixed same-page evaluation set. Up to five disjoint samples were drawn at each budget, depending on the size of the training pool, and each sample was used in five training runs. Every run received 300 optimizer steps.

Table 1: Generalized-model performance.

Model	Params. (M)	CER		WER	
		Micro	Macro	Micro	Macro
kraken CRNN	4.64	10.84	12.41	34.63	39.23
PP-OCRv6 tiny	0.68	14.25	15.24	43.52	47.35
PP-OCRv6 small	3.22	9.33	9.80	30.38	33.91
PP-OCRv6 medium	15.86	7.03	7.15	23.36	26.24

¹ Model weights and a full corpus inventory are available on Zenodo <https://doi.org/10.5281/zenodo.21788403>; <https://doi.org/10.5281/zenodo.21788405>; and <https://doi.org/10.5281/zenodo.21788410>.

² The test set is available on Zenodo at <https://zenodo.org/records/anonymized>.

In all experiments, CER and WER are computed after NFD normalization and the normalization of all whitespace to U+0020. Runtime is measured on identical line images using an NVIDIA A40 and an Intel Xeon Silver 4316.

Table 2: Architecture adaptation ablation (task 1 macro CER, %).

Variant	48 px	96 px	96 px + AdamW/Muon
PP-OCRv6 tiny	24.05	12.57	13.13
PP-OCRv6 small	19.28	10.54	10.50
PP-OCRv6 medium	18.35	10.10	9.77

4 Analysis

4.1 Generalized models

The generalized models are first evaluated on the held-out 33-language pretraining test set. As shown in Table 1, PP-OCRv6 small and medium outperform the kraken baseline in both character and word error rate, whereas the tiny variant performs worse. The progression of error rates from tiny to medium is substantial, reducing macro CER from 15.2% to 7.2%, which indicates that model capacity is the primary limitation when learning from a highly heterogeneous corpus.

4.2 CATMuS, CMMHWR, and CoMMA

The effects of the adaptations to PP-OCRv6 explained above are shown in Table 2. As removal of confidence-window filtering and fixed-width resizing is a prerequisite for historical ATR, only the line height and optimizer change were ablated.

Increasing the line height accounts for almost all of the improvement, reducing CMMHWR task 1 macro CER by 8.3-11.5 percentage points. Muon with AdamW was initially adopted to accelerate convergence, but produces scores broadly in line with pure Adam training. Adam training at 96 pixels was also found to be less stable, whereas no corresponding failures were observed with the hybrid optimizer. The 96-pixel configuration with Muon and AdamW is therefore used for the remaining PP-OCRv6 experiments.

The full-corpus from-scratch results are reported in Table 3. The kraken CRNN obtains lower CER and WER on task 1, task 2, and CoMMA. PP-OCRv6 small and medium have marginally lower CER on Czech, but not lower WER.

The effect of pretraining the generalized models and subsequent CATMuS fine-tuning is shown in Figure 2. The isolated markers give the scores of models trained from scratch on the complete corpus;

Table 3: CATMuS from-scratch results (CER; WER in parentheses, %).

Test set	kraken CRNN	PP-OCRv6		
		Tiny	Small	Medium
<i>CMMHWR</i>				
Task 1: French	4.32 (18.52)	7.34 (31.23)	5.39 (23.94)	4.94 (22.22)
Task 1: Latin	11.00 (44.92)	16.41 (62.00)	13.32 (54.06)	12.17 (50.98)
Task 1: Spanish	11.85 (39.24)	15.65 (49.35)	12.79 (42.57)	12.21 (42.17)
Task 2: Occitan	8.69 (39.55)	12.04 (51.87)	9.70 (44.82)	9.30 (44.31)
Task 3: Czech	26.67 (79.76)	28.16 (81.61)	25.93 (80.56)	25.99 (80.33)
<i>CoMMA</i>				
Latin	12.77 (45.98)	17.23 (59.79)	14.07 (52.49)	13.54 (51.39)
French	22.63 (55.79)	27.62 (65.40)	24.05 (58.98)	24.31 (58.59)

the connected series begin with the zero-shot generalized models and continue over the five fine-tuning tiers; the dashed line gives the Medusa result. Fine-tuning on only 25% of CATMuS improves every PP-OCRv6 variant over its counterpart trained from scratch on the complete corpus across all seven test sets. More fine-tuning data generally reduces error on task 1, task 2, and CoMMA. On Czech, however, every generalized model

performs best zero-shot and performance tends to deteriorate as more CATMuS data are introduced. PP-OCRv6 medium also outperforms Medusa at this point, with 12.4% CER against 15.7%, before CATMuS fine-tuning reverses the ordering. This indicates forgetting of cross-family knowledge acquired during pretraining.

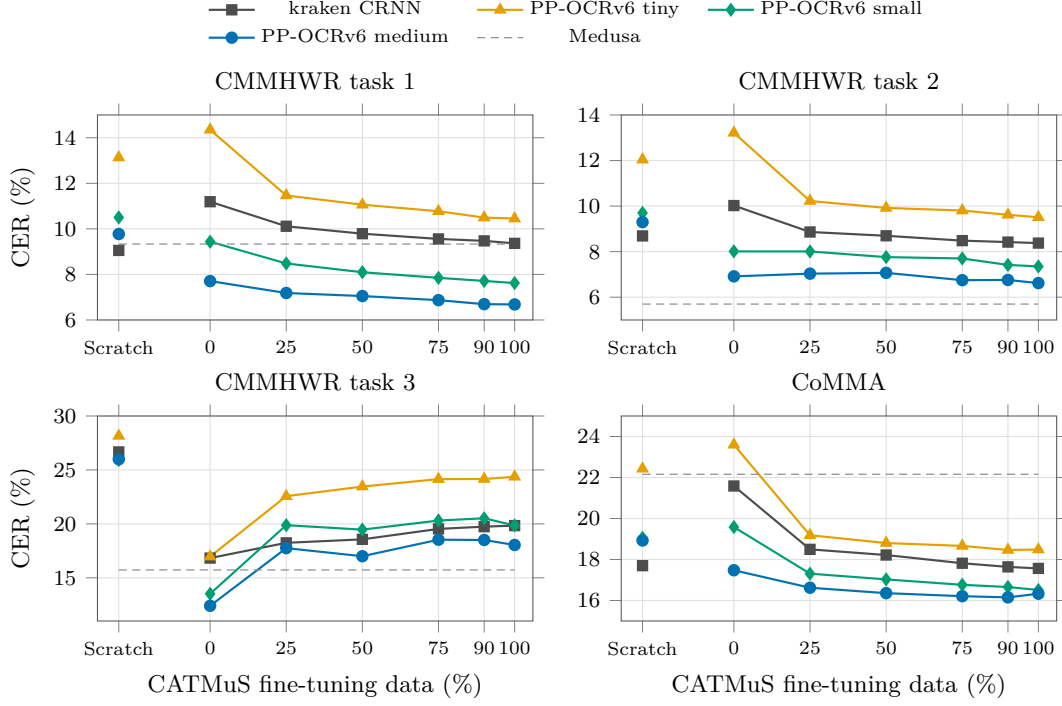


Figure 2: CATMuS training and fine-tuning by evaluation set.

Table 4 compares the fully fine-tuned generalized models with Medusa. PP-OCRv6 medium obtains lower CER and WER on task 1 and CoMMA, while Medusa remains better on Occitan and Czech. The generalized-model pretraining corpus contains no Occitan and only minuscule amounts of Czech, while CATMuS contains neither language. Medusa, in contrast, was trained on extensive Occitan and Czech material [22], explaining its better scores on tasks 2 and 3.

Table 4: CATMuS-adapted models and Medusa (CER; WER in parentheses, %).

Test set	kraken CRNN	PP-OCRv6			Medusa ^a
		Tiny	Small	Medium	
<i>CMMHWR</i>					
Task 1: French	5.01 (21.29)	6.26 (27.52)	4.07 (18.69)	3.36 (15.81)	3.74 (15.91)
Task 1: Latin	12.61 (51.03)	13.19 (54.22)	10.05 (44.26)	8.70 (39.67)	13.07 (43.96)
Task 1: Spanish	10.47 (38.03)	11.91 (41.50)	8.74 (33.90)	7.99 (31.02)	11.19 (32.74)
Task 2: Occitan	8.38 (39.23)	9.51 (44.42)	7.35 (38.57)	6.62 (36.03)	5.70 (21.91)
Task 3: Czech	19.84 (72.15)	24.37 (78.92)	19.86 (72.20)	18.05 (70.29)	15.73 (64.94)
<i>CoMMA</i>					
Latin	13.48 (50.89)	13.83 (53.19)	11.52 (45.36)	11.30 (42.60)	16.08 (45.54)
French	21.65 (54.44)	23.14 (58.66)	21.50 (53.50)	21.36 (51.39)	28.24 (55.74)

^a The task 1-3 CERs of 8.03%, 5.24%, and 10.8% reported for Medusa [22] could not be reproduced with the released DocWorkflow pipeline. The values shown here were obtained with that pipeline and its documented configuration.

4.3 Few-shot adaptation

The few-shot results are shown in Figure 3. A budget of zero indicates zero-shot accuracy of the generalized model. Relative to this starting point, the kraken baseline improves with only 20 training lines on all four manuscripts. PP-OCRv6 instead performs worse after 20 lines, improves after 40 lines on one manuscript and after 80 lines on two others, and does not improve within the maximum available budget of 80 lines on the fourth. This provides limited evidence that the recurrent baseline is more effective in the extreme few-shot regime. The slower improvement of PP-OCRv6 is offset by its lower absolute CER, which is below that of the baseline on all four manuscripts from 40 training lines onwards.

Variation across data draws and training runs is small: with one exception, the combined standard deviation remains below 0.8 percentage points and falls below 0.4 percentage points from 80 training lines onwards. PP-OCRv6 is nevertheless somewhat more variable than the baseline at the smallest budgets, where it does not improve upon the base model.

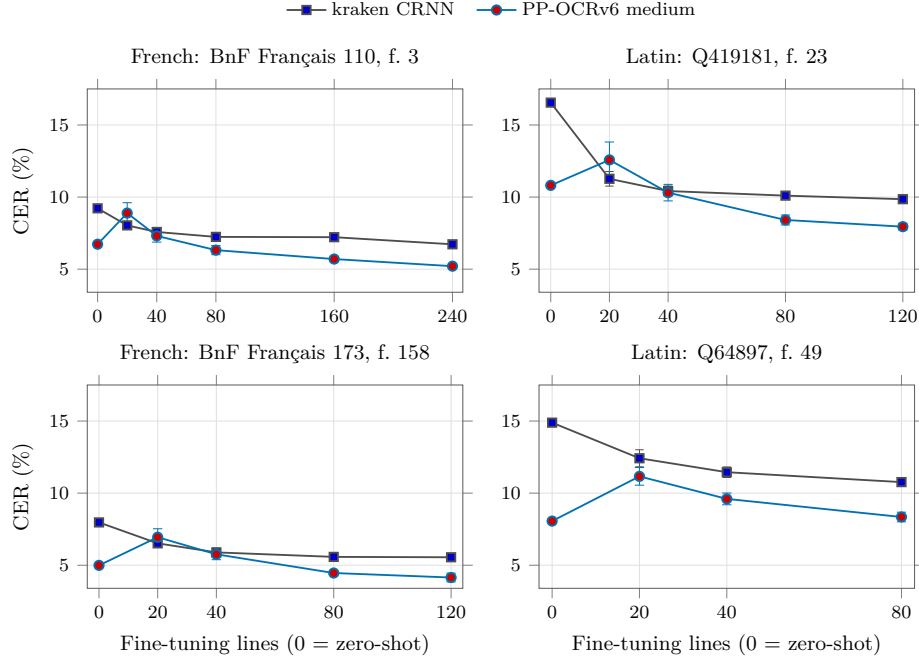


Figure 3: Few-shot CER by manuscript and training-set size.

4.4 Arabic-script transfer

Table 5 reports the MAKHZAN results by language and as an unweighted macro average, reproducing the model ranking observed on the Latin-script evaluations. PP-OCRv6 small and medium outperform the kraken baseline in macro CER and WER both zero-shot and after fine-tuning, while the tiny model does not. PP-OCRv6 medium is already the strongest model zero-shot, and its macro CER increases slightly from 18.0% to 18.8% after fine-tuning. The persistence of this performance gap on Arabic-script material makes it unlikely that the Latin-script results arise merely from an interaction between the larger volume of Latin-script pretraining data and the greater effective capacity of PP-OCRv6.

Table 5: MAKHZAN recognition results (CER; WER in parentheses, %).

Evaluation	kraken CRNN	PP-OCRv6		
		Tiny	Small	Medium
<i>Generalized models, zero-shot</i>				
Arabic	24.46 (70.97)	28.16 (81.57)	20.67 (66.02)	16.05 (52.95)
Persian	36.42 (86.84)	39.19 (90.08)	30.35 (80.53)	24.17 (70.32)
Ottoman Turkish	20.21 (67.88)	22.57 (73.72)	17.84 (64.72)	13.76 (53.91)
Macro	27.03 (75.23)	29.98 (81.79)	22.95 (70.43)	17.99 (59.06)
<i>Fine-tuned on MAKHZAN</i>				
Arabic	22.23 (66.61)	28.66 (76.37)	25.37 (70.80)	23.63 (65.94)
Persian	34.20 (84.32)	33.60 (84.15)	27.71 (75.88)	23.01 (67.52)
Ottoman Turkish	17.15 (61.88)	15.74 (58.73)	11.86 (48.61)	9.83 (41.65)
Macro	24.52 (70.94)	26.00 (73.08)	21.65 (65.09)	18.82 (58.37)

4.5 Inference speed

With PP-OCRv6 small, the improvement in recognition accuracy does not incur a substantial reduction in throughput on GPU. Its generalized model reduces macro CER from 12.4 to 9.8% and it outperforms the kraken CRNN on almost all test sets after CATMuS fine-tuning. At the same time, it is faster up to batch size 16 and remains close at batch 32, as shown in Figure 4, processing approximately 610 instead of 640 lines per second.

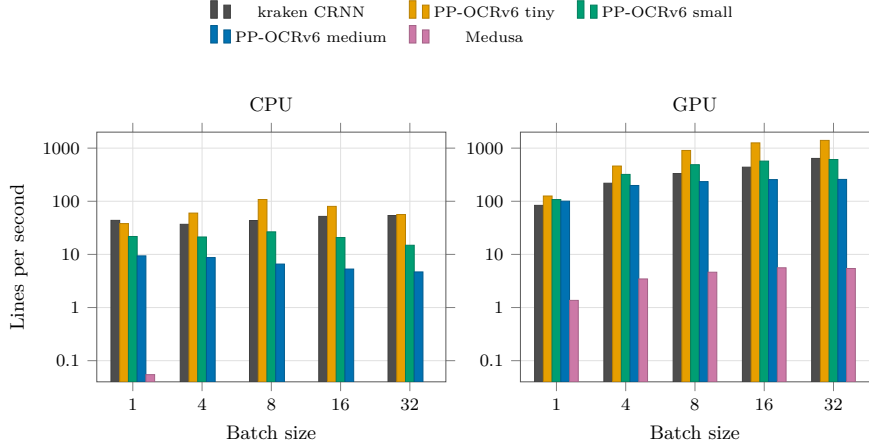


Figure 4: Inference throughput by batch size.

The results for tiny and medium reflect their respective model capacities. Tiny reaches approximately 1400 lines per second but is less accurate than the CRNN. Medium, which achieves the lowest overall error rates, is faster than the baseline only with batch size 1 and then levels off at approximately 260 lines per second. On the CPU, only tiny exceeds the CRNN. Even medium nevertheless remains approximately 45 times faster than Medusa.

The A40 uses NVIDIA’s Ampere architecture; Hopper and Blackwell hardware and software optimizations are likely to increase PP-OCRv6 throughput relative to the recurrent baseline.

5 Conclusion

PP-OCRv6 can be adapted to historical ATR without changing its basic architecture or substantially increasing its capacity. The main changes are an increased input line height, proportional line widths, and the omission of confidence-window filtering. After these adaptations, from-scratch training on CATMuS is competitive with the kraken CRNN, while generalized-model pretraining yields markedly better generalization. The medium model reduces macro CER on the 33-language test set from 12.4 to 7.2% and also transfers better to Arabic-script material. Corpus-specific fine-tuning further gives lower error rates than Medusa on CMMHWR task 1 and CoMMA. Only extreme few-shot adaptation remains less consistent, with PP-OCRv6 generally requiring 40-80 lines before improving on the base model.

The adapted models preserve CTC decoding and add no language-generative mechanism associated with hallucination. The small variant most clearly offers the free lunch: lower generalized and fine-tuned error rates, fewer parameters, and comparable or higher GPU throughput than the CRNN. The medium variant improves accuracy further at the cost of peak batched throughput, but remains far less expensive than a large VLM.

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